Engg. Mechanics.

Mechanics: The science which deals with the states of rest & motion of bodies under the influence of forces.

rest → Statics

motion → dynamics.

Body: Mass + space {Volume} = Matter

Mass = Body

Anything which has a property of Inertia is known as body.

Inertia: Resistance shown by a body against accn.

The standard def. of Inertia is obtained from Newtons' 1st Law {Law of Inertia}

The resistance shown by a body against any kind of change in its states of

1. Rest $\{\vec{V} = 0\}$
2. Uniform Motion along st. line. $\{\vec{V} = c\}$

How many types of Inertia does a body have?

There are 2 types of Inertias since there are 2 types of accn.

1. Linear Inertia {Mass of the body} $\vec{F} = m.\vec{a}$
2. Rotational Inertia {Mass M.O.I.} $\vec{T} = I.\vec{\alpha}$

Note: Area M.O.I. is not a property of body. It is a property of C/s.

Body → Rigid Body {Non Deformable} (E.M.) {E.M.}
Body → Deformable Body → Solid {S.O.M.}
Deformable Body → Fluid. {F.M.}

Rigid Body: A body in which the distance b/w any two particles doesn't change upon the application of any amount of load.

Strain = 0

Any elastic Modulus → ∞

Particle: A body whose dimensions are negligible w.r.t. the frame of ref. is a particle.

* Particle can observe only translational DOF
* Rotational Inertia abt. its own axis is zero.

Syllabus

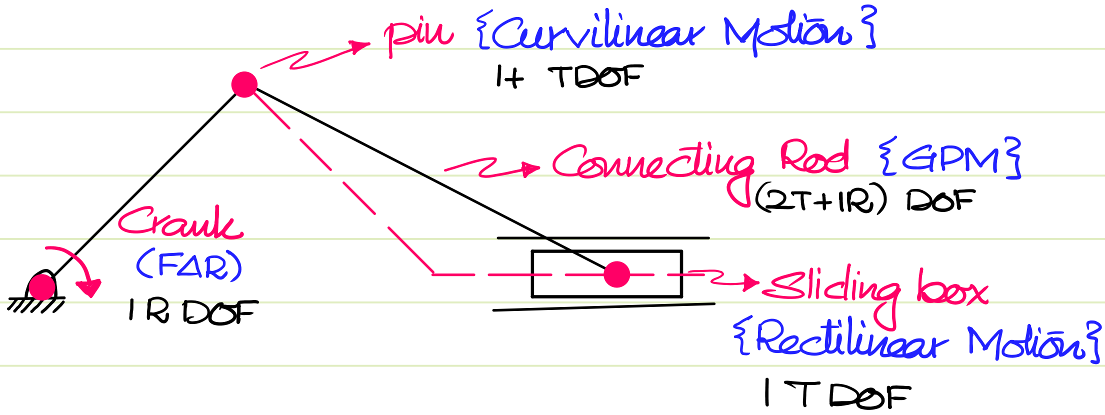
A. Statics

1. System of forces & Moments.
2. Equilibrium***
   - Introduction
   - Analysis of Trusses & Frames.
   - Friction
   - Virtual Work

B. Dynamics

3. Kinematics
   - 3.1. Particle Study.
     - * Rectilinear Motion
     - * Curvilinear Motion
   - 3.2. Rigid Body St.
     - * Fixed Axis Rot
     - * General Plane Motion.

4. Kinetics
   - 4.1. Newton's 2nd Law of Motion
   - 4.2. Work-Energy Th^rm.
   - 4.3. Lagrangian.
   - 4.4. Impulse-Mom Th^rm.



Weightage :

GATE : 4M - 6M.

ESE {prelims} : 30 - 50

ESE {Mains} : 20 Marks.

Ref. Books :

1. Beer & Johnston
2. F.L. Singer & Pytel
3. R.C. Hibbler
4. Timoshenko & Young.

1. System of Forces & Moments

Force: Anything which changes or tries to change a body's state of rest or uniform motion along st. line.

$\Rightarrow$ Force leads to $\Delta\vec{V}$ $\Rightarrow$ force leads to acceleration.

$$\boxed{\vec{F} = m.\vec{a}}$$

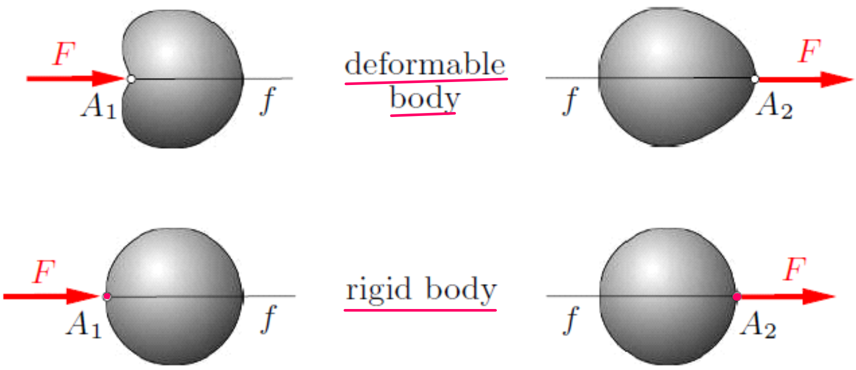
S.I.: $kg.m/s^2$ {Newton}

CGS: $g.cm/s^2$ {Dyne $= 10^{-5}N$}

In order to identify the effect of force on a body, the following 3 ideas about force are required.

1. Magnitude of force
2. Direction of force
3. pt. of app. or line of action.

Principle of Transmissibility: The net effect of a force on a rigid body remains unchanged when it is represented anywhere along its line of action in the same direction.



This law is very important in the calculation of Moment of force.

Systems of forces: A set of forces is known as system of forces.

A. Based on the plane in which forces are:

1. Coplanar SOF: all forces in common plane {2D}
2. Non Coplanar SOF: all forces are not in common plane {3D}

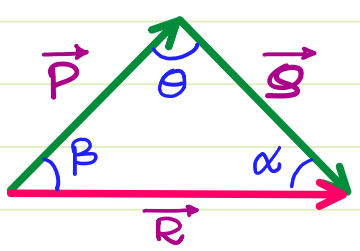
B. Based on the Concurrency of lines of action

1. Concurrent SOF: All the lines of action coincide at a common point.
2. Non Concurrent SOF: The lines of action donot coincide at a common pt.

Resultant of forces: A single force which can replace a system of forces while creating the same translational & rotational effect is known as resultant of forces.

There are 3 laws corresponding to resultant of forces.

1. Δ^{le} Law: {2 forces}

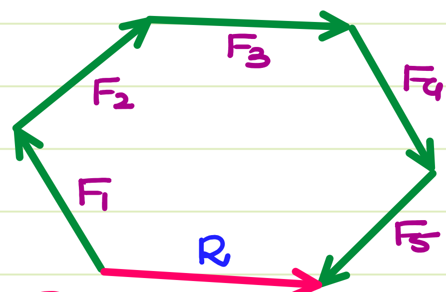


Cosine Rule: $R^2 = P^2 + Q^2 - 2PQ\cos\theta$

Sine Rule: $\dfrac{P}{\sin\alpha} = \dfrac{Q}{\sin\beta} = \dfrac{R}{\sin\theta}$

2. Polygon Law: {2+ forces}

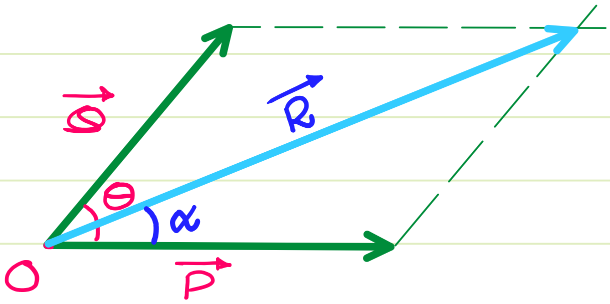
$\sim$ to Δ^{le} Law



Polygon law is obtained by repeating Δ^{le} law

Abdul Bari - 9573745717

3. Parallelogram Law : {2 forces such that tails coincide}



$$R^2 = P^2 + Q^2 + 2PQ\cos\theta$$

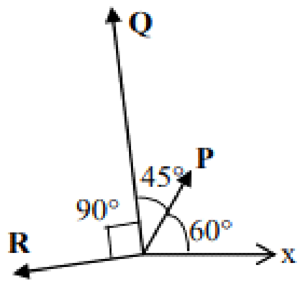
$$\alpha = \tan^{-1}\left\{\frac{Q\sin\theta}{P + Q\cos\theta}\right\}$$

Note: If there are more than 2 coplanar forces present, then the forces can be converted into 2 Mutually ⊥ar forces $\{\Sigma F_x$ & $\Sigma F_y\}$

$$R = \sqrt{\{\Sigma F_x\}^2 + \{\Sigma F_y\}^2}$$

2M CE

The magnitudes of vectors **P, Q** and **R** are 100 kN, 250 kN and 150 kN, respectively as shown in the figure.



The respective values of the magnitude (in kN) and the direction (with respect to the x-axis) of the resultant vector are

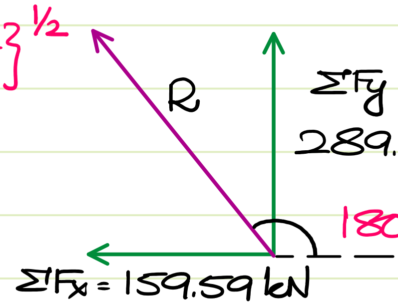
(A) 290.9 and 96.0° (B) 368.1 and 94.7° (C) 330.4 and 118.9° (D) 400.1 and 113.5°

$\Sigma F_x = P\cos 60° + Q\cos 105° + R\cos 195° = -159.59$ kN.

$\Sigma F_y = P\sin 60° + Q\sin 105° + R\sin 195° = 289.26$ kN.

$R = \{(-159.59)^2 + (289.26)^2\}^{1/2}$

$= 330.36$ kN.

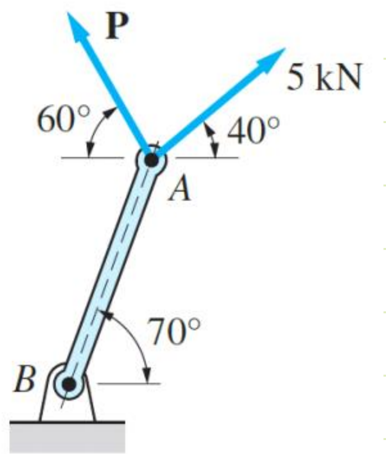


$180° - \tan^{-1}\left\{\frac{289.26}{159.59}\right\}$

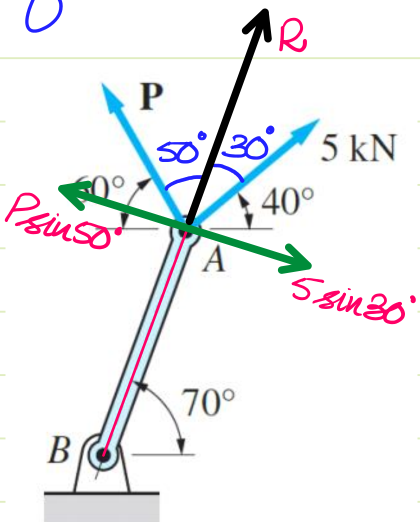
$= 118.88°$

Abdul Bari - 9573745717

The two forces shown act on the structural member **AB**. Determine the magnitude of ***P*** such that the resultant of these forces is directed along **AB**.



If the resultant is along AB ⇒ the net force ⊥ar to AB = 0

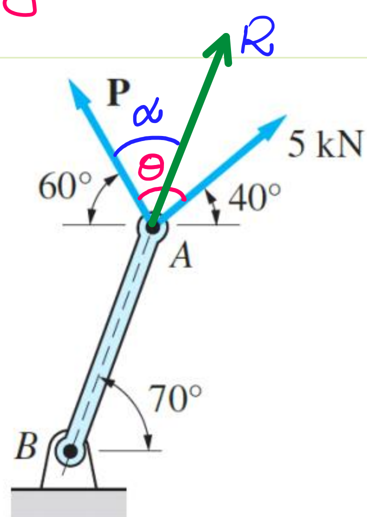


$5\sin 30° = P\sin 50°$

$\Rightarrow P = \dfrac{2.5}{\sin 50°} = 3.2635\,kN.$

$R = 5\cos 30° + P\cos 50° = 6.4278\,kN.$

* By ||gm Law



$\theta = 80°\,;\ \alpha = 50°\,;\ Q = 5kN\ ;\ P = ?$

$\tan\alpha = \dfrac{Q.\sin\theta}{P + Q\cos\theta}$

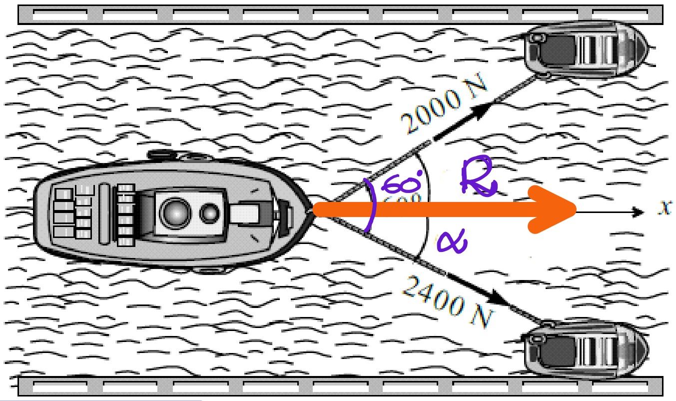
$\Rightarrow P = \dfrac{Q\sin\theta}{\tan\alpha} - Q\cos\theta$

$P = 3.2635\,kN.$

$R = \{P^2 + Q^2 + 2PQ\cos\theta\}^{1/2}$

$R = 6.4278\,kN.$

Two locomotive on opposite banks of a canal pull a vessel moving parallel to the banks by means of two horizontal ropes. The tension in these ropes are **2000 N** and **2400 N** while the angle between them is **60^0**. Find the resultant pull on the vessel and the angle between each of the ropes and the sides of the canal.

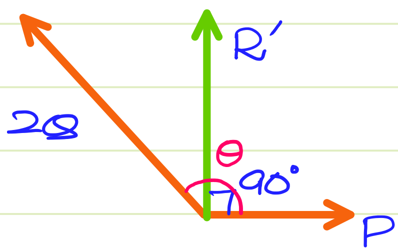


$$R = \sqrt{2000^2 + 2400^2 + 2(2000)(2400)\cos 60^\circ}$$

$$R = 3815.75\,N.$$

$$\alpha = \tan^{-1}\left\{\frac{2000 \sin 60^\circ}{2400 + 2000\cos 60^\circ}\right\} \qquad \alpha = 27^\circ$$

$$60^\circ - \alpha = 33^\circ$$



02. The resultant of two forces P and Q is R. If Q is doubled, the new resultant is perpendicular to P. Then,

(a) P = R (b) Q = R

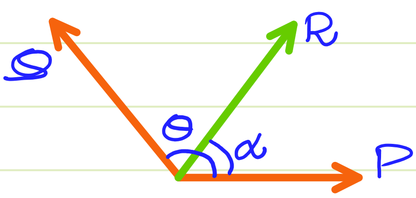
(c) P = Q (d) None of the above

$\Sigma F_x = 0$ {∵ R′ is in y axis}

$\Rightarrow P = 2Q\cos\{180^\circ - \theta\}$

$\Rightarrow \boxed{P = -2Q\cos\theta}$

Initial Case:



By ||gm law:

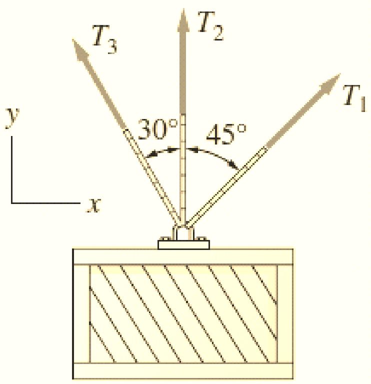
$$R^2 = P^2 + Q^2 + 2PQ\cos\theta$$

$$\Rightarrow R^2 = \{-2Q\cos\theta\}^2 + Q^2 + 2\{-2Q\cos\theta\}\cos\theta$$

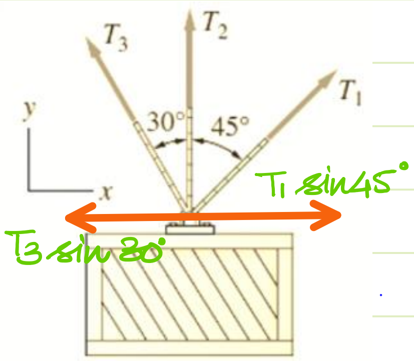
$$\Rightarrow R^2 = \cancel{4Q^2\cos^2\theta} + Q^2 - \cancel{4Q^2\cos^2\theta}$$

$$R^2 = Q^2$$

Abdul Bari - 9573745717



The resultant force of the three cable tensions that support the crate is $\mathbf{R} = 500\mathbf{j}$ N. Find T_1 and T_3, given that $T_2 = 300$ N.



$\vec{R} = \{0\hat{i} + 500\hat{j}\}$ N.

$T_2 = 300\hat{j}$ N

$\because \sum F_x = 0$

$T_1 \sin 45° = T_3 \sin 30°$

$\boxed{T_1 = T_3/\sqrt{2}}$

$\sum F_y = 500 = T_1 \cos 45° + T_2 + T_3 \cos 30°$

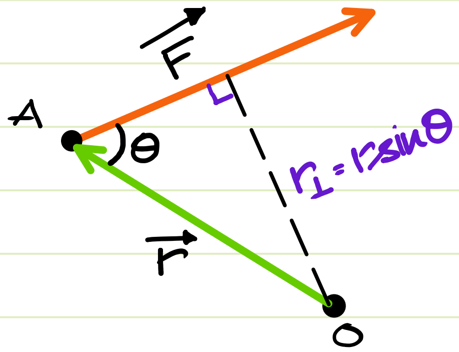
$\Rightarrow 500 = \frac{T_3}{2} + 300 + T_3\frac{\sqrt{3}}{2}$

$\Rightarrow T_3 = \frac{400}{1+\sqrt{3}}$

$\therefore$ $\boxed{\begin{array}{l} T_3 = 146.41\text{ N} \\ T_1 = 103.53\text{N.} \end{array}}$

Abdul Bari - 9573745717

Moment of Force : The rotational effect of a force about a particular point or an axis is known as Moment of force {Vector Quantity}



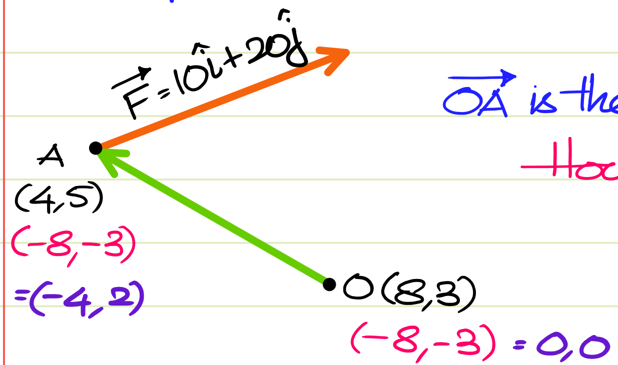
$$\vec{M} = \vec{r} \times \vec{F} \quad \{\vec{r} = \vec{OA}\}$$

$$= F.r.\sin\theta$$

$\hookrightarrow r_{\perp}$ {Lever/Moment arm}

$$\Rightarrow \bar{M} = F.r_{\perp} \qquad \{SI : N.m\}$$

Q: Find the moment of force $\vec{F} = \{10\hat{i} + 20\hat{j}\}$ N acting at pt A:(4,5) about pt O:(8,3) {All lengths are in m}.



$\vec{OA}$ is the position of A w.r.t O as origin

~~How to create O as origin~~

$$\vec{r} = -4\hat{i} + 2\hat{j}$$

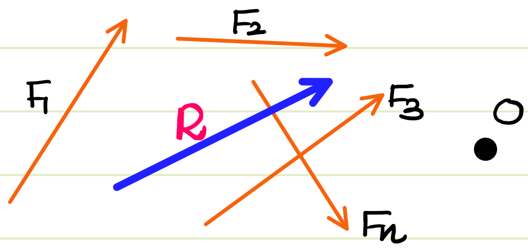
$$\vec{M}_O^F = \{-4\hat{i} + 2\hat{j}\} \times \{10\hat{i} + 20\hat{j}\}$$

$$= -80\hat{k} - 20\hat{k}$$

$$\therefore \vec{M}_O^F = -100\hat{k} \text{ N.m}$$

–ve : clockwise

Varignons' Th^rm : The sum of moments of forces in a system must be equal to Moment of Resultant.



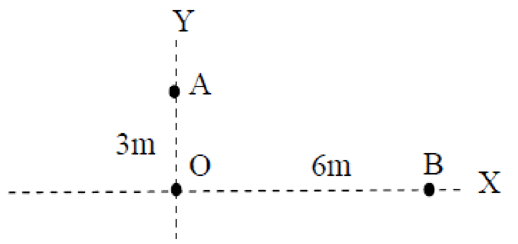
$$\boxed{\sum_{i=1}^{n} M_O^{F_i} = M_O^{R}}$$

Note : This th^rm is helpful in finding the location of resultant of forces.

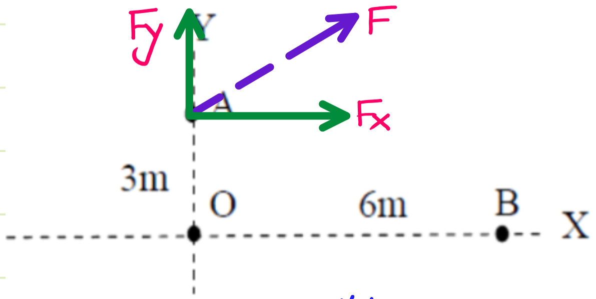
Abdul Bari - 9573745717

Note: If the moment of a non-zero force is zero about a point, then the line of action of force must pass thru that point $\{r_\perp = 0\}$.

03. With reference to the figure shown below, the moment due to a certain force, F, is 180 N-m clockwise about the origin O and 90 N-m anticlockwise about the point B. If the moment due to the same force, F about A is zero then the value of the force F is



(a) 20 N (b) 75 N (c) 100 N (d) 125 N



By Varignon's th'm:

$M^F = M^{F_x} + M^{F_y}$

1. Moment abt pt "O": $M_O^F = M_O^{F_x} + \cancel{M_O^{F_y}}$

$-180 = -F_x\{3\} \Rightarrow \boxed{F_x = 60N.}$

2. Moment abt pt "B": $M_B^F = M_B^{F_x} + M_B^{F_y}$

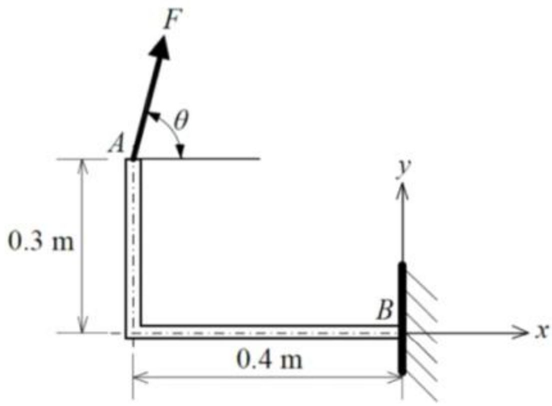
$90 = -60\{3\} - F_y\{6\}$

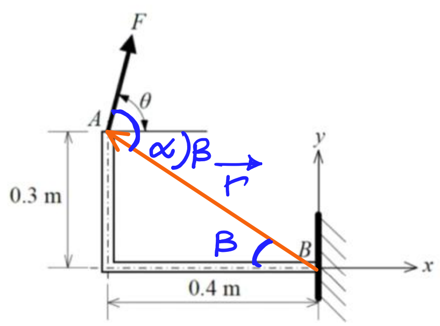
$\Rightarrow 6F_y = -270 \Rightarrow \boxed{F_y = -45N.}$

$F = \sqrt{F_x^2 + F_y^2} \Rightarrow \boxed{F = 75N.}$

IM, XE

A rigid and thin L-shaped bracket is fixed to the wall at point B, and a force F is applied at point A as shown. For a given force F, the point B experiences the **maximum clockwise moment** when the inclination *(in degrees)* with the x-axis is..........*(up to two decimal places)*.





$M = F.r.\sin\alpha$

for Max Moment, $\sin\alpha$ must be Max

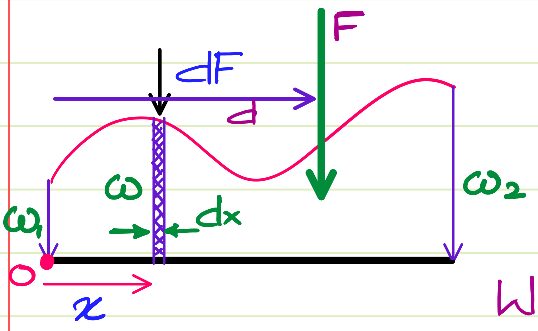
$\Rightarrow \alpha = 90°$ $\therefore \theta = 90° - \beta$

$= 90° - \tan^{-1}\{0.3/0.4\}$

$\therefore \theta = 53.13°$

Abdul Bari - 9573745717

Distributed Loading: ω : Intensity of loading $\left\{\dfrac{\text{load}}{\text{length}}\right\}$



$\omega = f(x)$ variable

$dF = \omega . dx$

$F = \int \omega . dx$: Area under loading curve.

Where is force "F" concentrated?

Varignons' Th^rem: $dM_O = dF . x$

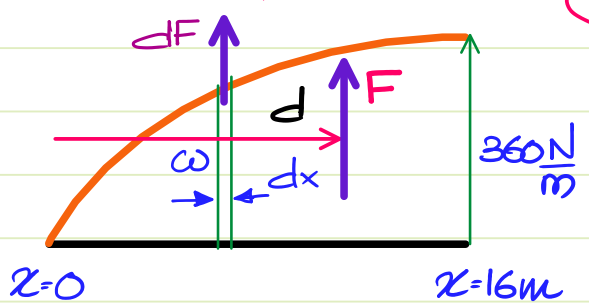
$M_O = \int dF . x = F . d$

$\Rightarrow d = \dfrac{1}{F} . \int dF . x$ {Centroid of the area under loading curve}

Given: The loading $\{\omega = 90x^{1/2}\}$ is a parabolic loading.

04. The 16 m wing of an aeroplane is subjected to a lift which varies from zero at the tip to 360 N/m at the fuselage according to $w = 90\sqrt{x}$ where x is measured from the tip. The resultant lift and its location from the wing tip are:

(a) 3840 N at 9.60 m (b) 4000 N at 11 m
(c) 3200 N at 8.50 m (d) 4500 N at 8 m



$dF = 90x^{1/2} . dx \Rightarrow F = \int_0^{16} 90x^{1/2} . dx = 60\{16\}^{3/2}$
$= 3840 N.$

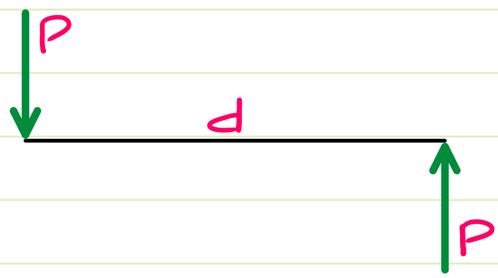
$d = \dfrac{1}{F}\int dF . x = \dfrac{1}{3840}\int_0^{16} 90x^{3/2} . dx = \dfrac{36 \times 16^{5/2}}{3840}$
$= 9.6 m$

F = 3840N at 9.6m from tip.

Abdul Bari - 9573745717

Couple: A special Moment created by a set of forces such that Resultant force = 0 but Net Moment is Non zero. $\{\Sigma F = 0 \text{ but } \Sigma \bar{M} \neq 0\}$.

The most primary couple is created by 2 equal & opposite forces seperated by an eccentricity.



$\Sigma F = P + \{-P\} = 0$

$\{\Sigma M = P.d\} \neq 0$

$\downarrow$

Couple $= C = P.d$ ↺

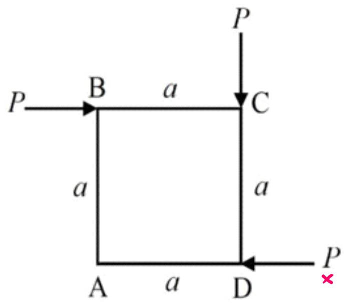
Properties of Couple:

1. Zero translational effect but Non zero rotational effect.
2. Couple is a free vector. Its value remains same when calculated about any point.
3. Equivalent force couple system {Many XE Qs}
   If a body experiences a set of forces & moments, then they can be replaced by a single force & couple abt any point.
4. Equilibrium.
   — A body experiencing couple can be maintained in eq^m only by a counter couple.

Abdul Bari - 9573745717

1M XE

A rigid square ABCD is subjected to planar forces at the corners as shown.



ΣF in the system = ΣF in eq^{ut} system

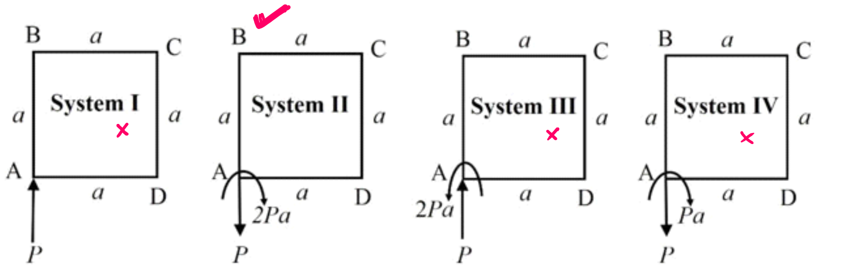
$0\hat{i} - P\hat{j}$ = ΣF eq^{ut} system.

ΣM_A in the system = ΣM_A in eq^{ut} system

$-Pa - Pa = -2Pa$

$2Pa$ ↻

For this planar force system, the equivalent force couple system at corner A can be represented as



2M, XE

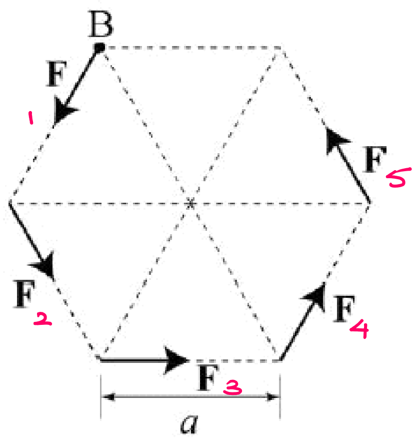
Consider the forces of magnitude **F** acting on the sides of the regular hexagon having side length a. At point B, the equivalent force and couple are, respectively,

ΣF = only F at ③

$\because F_1 = -F_4$

& $F_2 = -F_5$

$\therefore \Sigma F = F (\longrightarrow)$



ΣM_B must be anti cl.w. {by observation}

$\Sigma M_B = C$ by F_1 & F_4

$+ C$ by F_2 & F_5

$+ M_B^{F_3}$

$= 3\sqrt{3}\,F.a$ ↺

(A) F(⟵) and $3\sqrt{3}Fa$ (clockwise)

(B) F(⟶) and $\sqrt{3}Fa$ (clockwise)

(C) F(⟵) and $\sqrt{3}Fa$ (counter clockwise)

(D) F(⟶) and $3\sqrt{3}Fa$ (counter clockwise)

Abdul Bari - 9573745717

Q: 3 forces taken in sequence form a closed Δ^{le}. The magnitude of net moment abt. the centroid of the Δ^{le} is

a. half the area of Δ^{le}.
b. area of Δ^{le}
c. twice the area of Δ^{le} ✔
d. four times the area of Δ^{le}

Sol. ∵ The forces form a closed figure {Δ^{le}} ∴ the resultant {closing side} must be zero.

$$\Sigma F = 0 \quad \text{but} \quad \Sigma M \neq 0$$

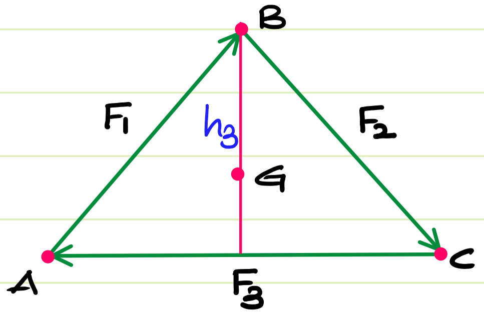
∴ ΣM = Couple.

∵ Couple is a free vector, ∴ $\Sigma M_G = \Sigma M_B$

$= F_3 \{h_3\}$

$= 2\{\frac{1}{2} F_3 . h_3\}$

$= 2 \times$ Area of Δ^{le}



2. Equilibrium

2.1. Fundamentals of Eq^m.

* Introduction
* Eq^{ns} of Eq^m.
* Newtons' 3rd Law of Motion
* Free Body Diagram {FBD}
* Concept of Reactions.

* Special Cases :
  1. Eq^m of 2 force Members.
  2. Eq^m of 3 force Members. {Lamis' Th^{rm}.}

Equilibrium : A state of a body in which the Net forces & Moments acting on the body are equal to Zero.

Eq^m $\neq$ Rest

$$\boxed{\begin{aligned} \Sigma\vec{F}=0 &\rightarrow \vec{a}=0 \\ \Sigma\vec{M}=0 &\rightarrow \vec{\alpha}=0 \end{aligned}}$$

Eq^m : $\vec{a}=0$

Rest : $\vec{V}=0$

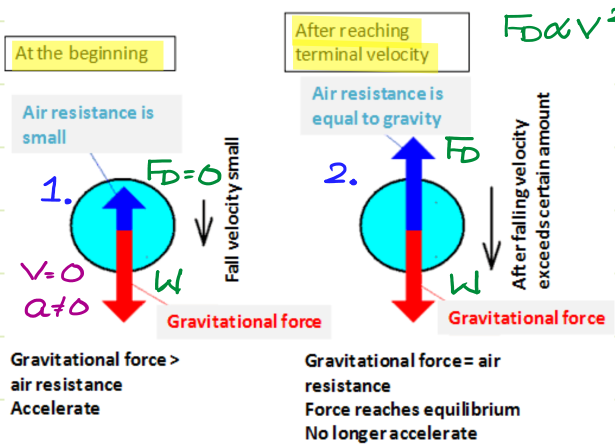
Eq^m & rest can exist regardless of each other.

* Temperory Rest : Rest w/o Eq^m $\{\bar{V}=0\ ;\ \vec{a}\neq 0\}$
* Permanent Rest : Rest with eq^m $\{\bar{V}=0\ ;\ \bar{a}=0\}$
* Uniform Velocity / Terminal Velocity Cond^n : $\bar{V}$ = const.
  Body has attained eq^m but it is in Motion $\{\bar{V}\neq 0\ ;\ \bar{a}=0\}$

$F_D \propto V^2$

1. $\bar{V}=0$ but $\bar{a}\neq 0$ {Temp. Rest}
   $\Sigma\vec{F}=W\downarrow$

2. $\Sigma F=0$ $\{F_D=W\}$ attained terminal velocity
   $V=c$ $\{a=0\}$



Abdul Bari - 9573745717

Eq^{ns} of Eq^{m}: $\Sigma\vec{F}=0$; $\Sigma\vec{M}=0$

1. Coplanar Concurrent SOF: $\Sigma F_x=0$; $\Sigma F_y=0$

2. Coplanar Non Concurrent SOF: $\Sigma F_x=0$; $\Sigma F_y=0$; $\Sigma M_z=0$

3. Non Coplanar Concurrent SOF: $\Sigma F_x=0$; $\Sigma F_y=0$; $\Sigma F_z=0$

4. Non Coplanar Non Concurrent SOF: $\Sigma F_x=0$; $\Sigma F_y=0$; $\Sigma F_z=0$
$\Sigma M_x=0$; $\Sigma M_y=0$; $\Sigma M_z=0$

*** Newtons' 3rd Law of Motion: For every action, there is an equal & opposite reaction, between 2 bodies in interaction.

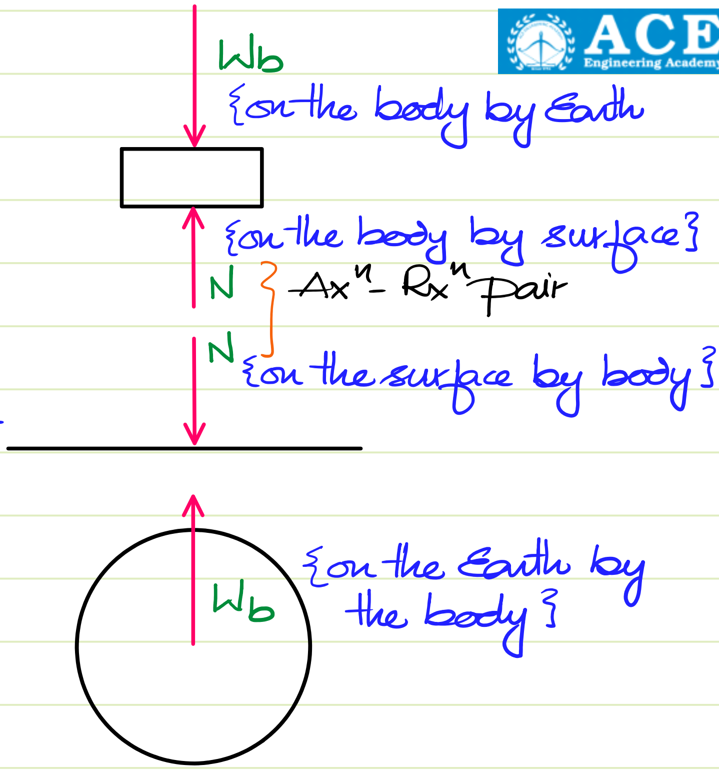
* ∴ Action - reaction pair can never be established on a single body. The pair must exist b/w the two interacting bodies.
* Ac^n & Reac^n must belong to the same family of force.
{If electromag. force is ac^n, then electromag. force must be rea^n}
* Ac^n & Reac^n have same line of action.
* Forces must exist in pairs.

** Free Body Diag^m {FBD}: A diag^m in which a body is isolated from its system & then all the forces acting on the body are shown.

* If ac^n & reac^n of a force is present within the system, then that force becomes an Internal force.

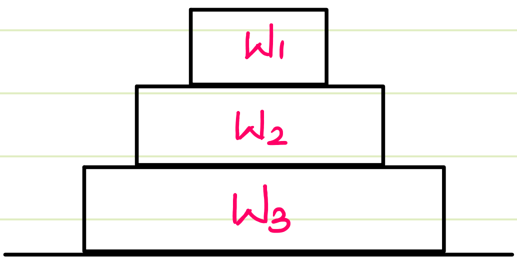
Abdul Bari - 9573745717

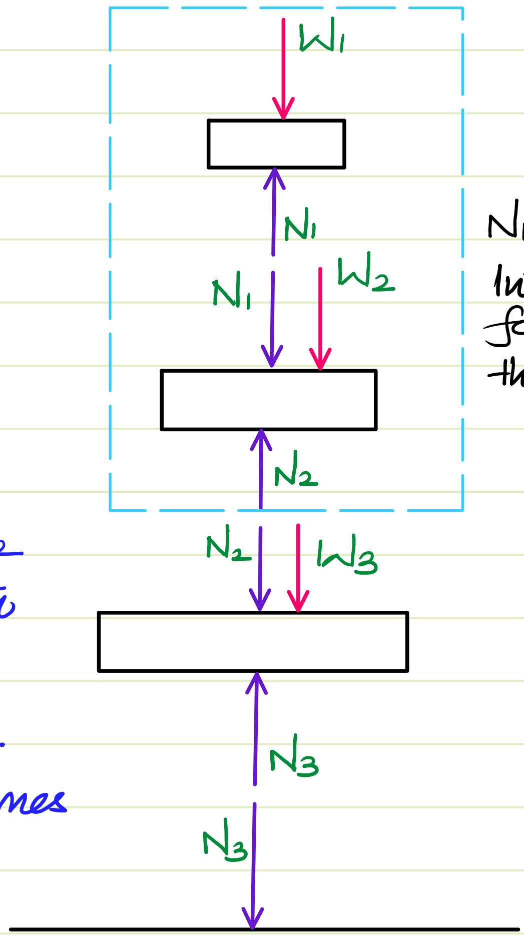
System



Wb & Wb forms an ax^n-reac^n pair
N & N forms an ax^n-reac^n pair.

* Draw FBD for the system shown

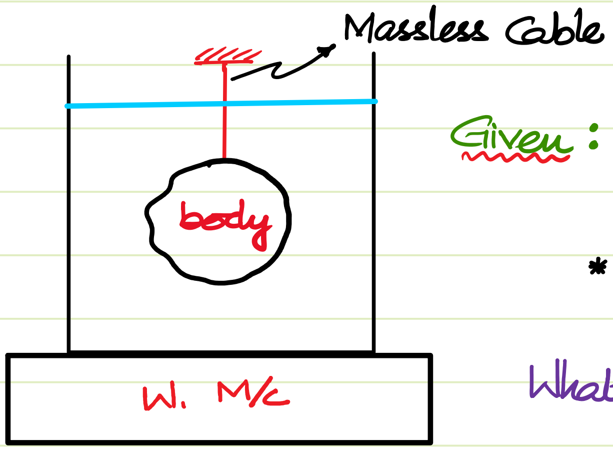




N$_1$ is an internal force for this system

If the combination of W$_1$ & W$_2$ is chosen as a system, then N$_1$'s ac^n & reac^n is present within the system.
∴ for that system N$_1$ becomes an internal force.

Abdul Bari - 9573745717
ACE Engineering Academy

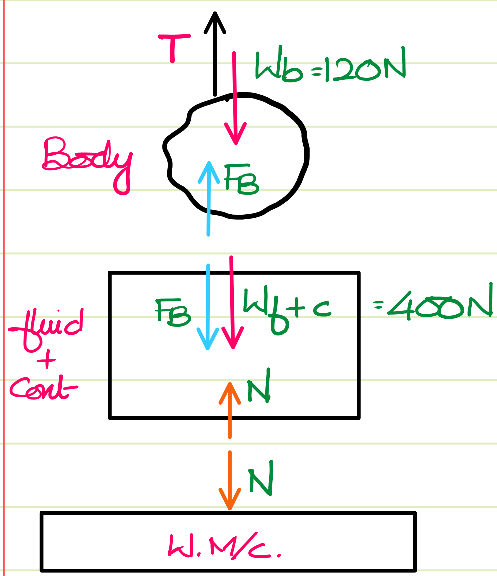


Given : * $W_{fluid + container} = 400N.$

* $W_{body} = 120N.$

* Buoyant force $= 45N.$

What is the reading on the W. M/c.



The reading on the W. M/c = Normal

By eq^n of eq^m for fluid + container

$$\Sigma F_y = 0$$

$$N - F_B - W_{f+c} = 0$$

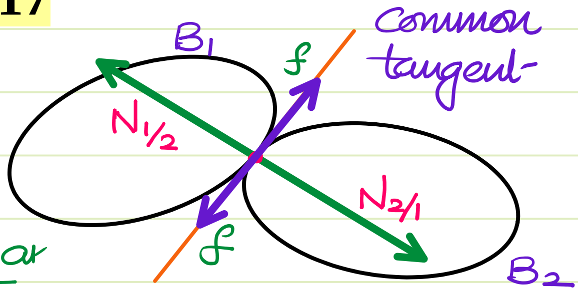
$$\Rightarrow N = F_B + W_{f+c}$$

$$N = 445N$$

* **Concept of Reaction:** When a body is supported by any support or Member, then an ac^n-reac^n pair is developed due to the support. When FBD of the body is drawn, the support must be released & the corresponding reac^n due to support must be shown on the body.

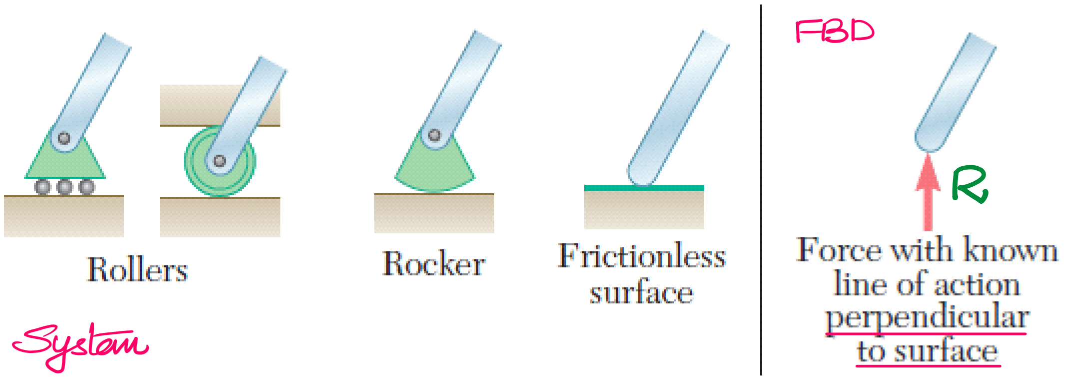
Whatever type of motion is resisted by the support, that type of reac^n is developed.

1. Contact Support :

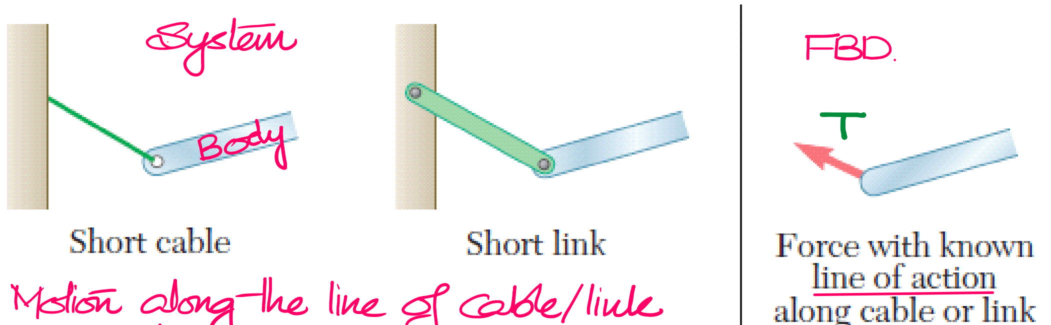


The motion of bodies ⊥ar to the common tangent is resisted.
∴ The acⁿ-reacⁿ pair must be developed in this dirⁿ.
These forces are known as Normal forces.
If the surfaces are rough & relative motion is trying to take place, then the Motion along the common tangent is resisted & these forces are known as friction forces.

2. Smooth Roller Support : { Similar to Normal force }



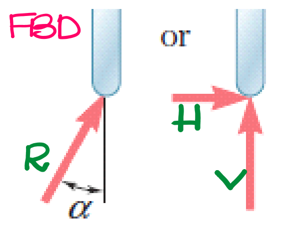
3. Massless Cable / Rope / Link :



Motion along the line of cable/link is resisted

4. Pin/Hinge Support : {Rotational Motion is allowed but translation is resisted}

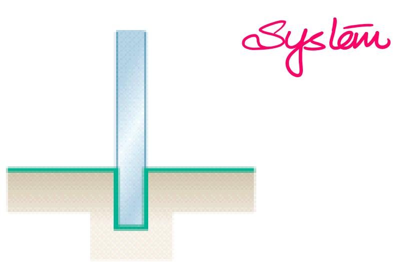
Frictionless pin or hinge — Rough surface



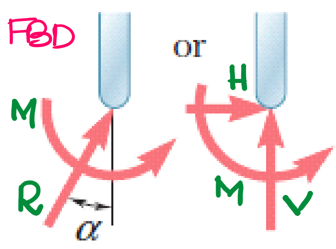
Force of unknown direction

2 unknowns :
2 unknown $\perp^{ar}$ forces (H & V)
or
1 unknown force is unknown dir^n. (R, α)

5. Fixed / Cantilever Support : {Translation & Rotation are resisted}



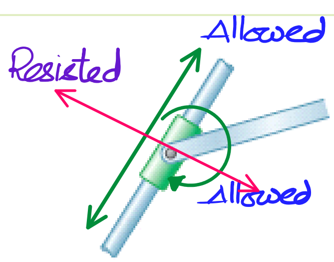
Fixed support



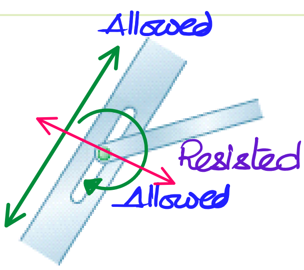
Force and couple

3 unknowns :
1. 2 Mutually $\perp^{ar}$ forces {H & V} & 1 Moment $reac^n$ {M}
2. 1 unknown force in unknown angle & Moment $reac^n$ {R, α, M}

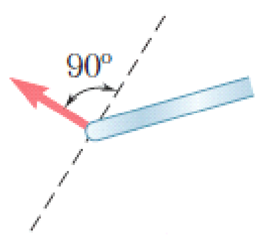
6. Collar / frictionless pin in a slot :



Collar on frictionless rod



Frictionless pin in slot



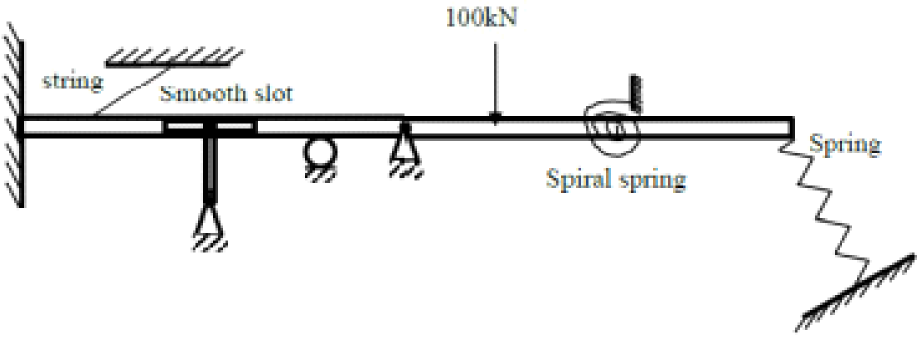
Force with known line of action perpendicular to rod or slot

Abdul Bari - 9573745717

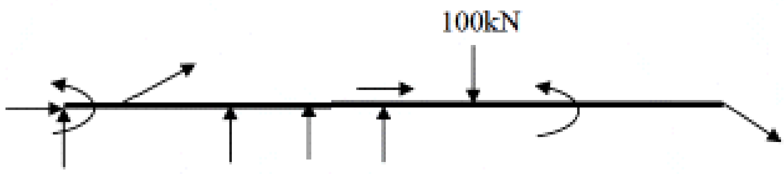
ACE Engineering Academy

IM XE

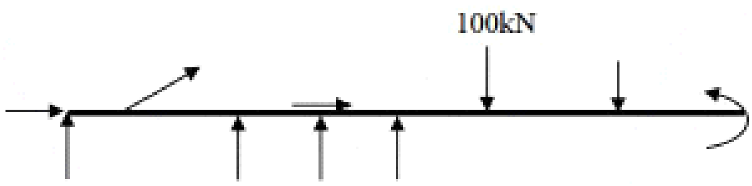
The figure shows a structure with supports. The correct free body diagram when the supports are removed is



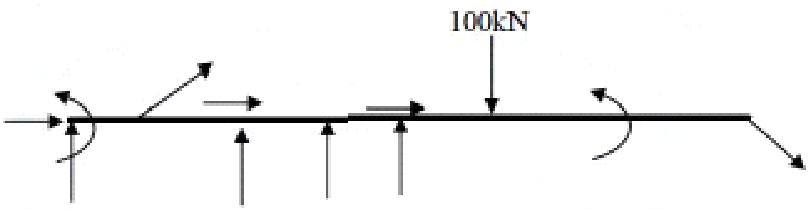
(A)



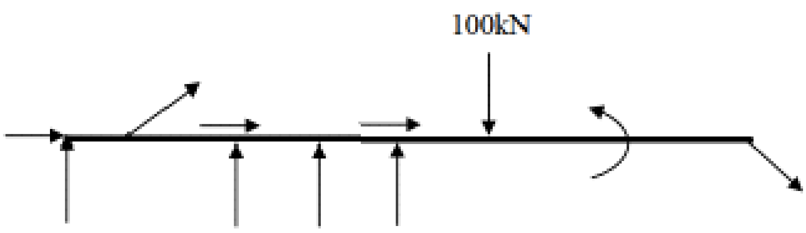
(B)



(C)



(D)



Abdul Bari - 9573745717